02-09-2026
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Thermodynamics

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Introduction

Thermodynamics was originally developed along with heat machines like the steam engine. The central point was that there are limits as to which extend thermal energy can be converted to work. Today thermodynamics has found use in practically all technical disciplines and involves much more than thermal engines.

On a shoestring and simplified, thermodynamics is about the thermal energy of stuff (thermo-) and its interaction with other stuff (-dynamics). The governing principle is that any macroscopic transfer of energy if governed by the likelihood of transfer on the molecular level. This likelihood is handled via the quantity *entropy* which quantifies the spreading of the energy.

Thermal energy is disordered (uncoordinated, without a common direction of possible action). Ordered energy is macroscopic mechanical energy and electrical energy in a wire (it acts, or it has the potential to act, in a common direction). This is spart of the story too, as the relevance of thermodynamics lies in its ability to describe the interplay between disordered and ordered energy.

Thermodynamics is also an abstraction treating the extreme cases, which are the ultimate limits for energy transfer. The term reversibility is central despite it is never realized in real systems. Argumentation is often supported by so called "*thermodynamic machines*" with are combinations of different ideal unit operations without friction or any other imperfections not given by the thermodynamics; imperfection that in theory can be minimized as much as we like, approaching zero.

In this context, thermodynamics is relevant for

- power generation from thermal energy sources.
- the function and efficiency of heat pumps.
- quantification of the energy content of a fuel available for a given application.

Basics of thermodynamics

The letter symbols

Different textbooks use either capital letter symbols or small cap symbols for the thermodynamic functions and other physical quantities. There seem to be no general agreement. Here capitals are used for absolute quantities (external quantities, depending on the size of the system) and small cap symbols for molar or mass-based quantities (internal quantities, independent on the size of the system). As an example, in this text "U" is used for internal energy (with the unit Joule, J) while "u" is used for molar or mass specific internal energy (J mol^{-1} or J kg^{-1}, respectively). Likewise, "C_V" is the absolute constant volume heat capacity (J K^{-1}) and "c_V" the molar or mass specific constant volume heat capacity (J mol^{-1} K^{-1} or J kg^{-1} K^{-1}).

The thermodynamic system

The "thing" we look at and assign quantities to is the *system* or the *thermodynamic system*. It constitutes a fixed volume, a fixed amount of substance, the interior of a box or something else with a

well-defined boundary. If the system is defined by walls, these walls may be included or excluded in the system. Everything outside the system (the rest of the universe) is called the surroundings. In thermodynamics, we look at changes in the system caused by exchange of energy or matter between the surroundings and the system. If energy enters the system, it is counted as positive, if it leaves, it is counted negative. Everything is from the perspective of the system. See Figure 1. This is rather trivial, yet important to note.

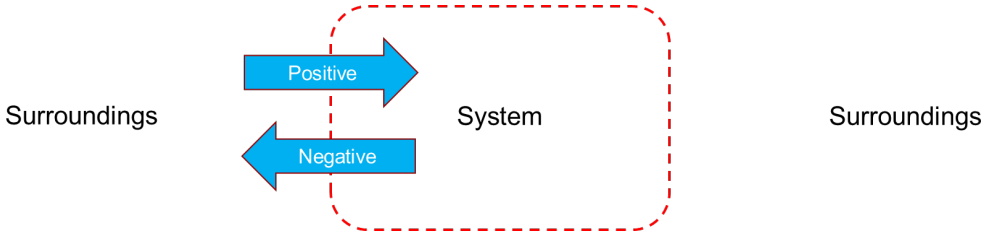


Figure 1. The thermodynamic system and its surroundings. The arrows represent flow of energy or matter.

Some more definitions and terms

- An *isolated* system is a system that does not exchange heat or matter with the surroundings.
- A *closed* system does not exchange matter with the surroundings.
- An *adiabatic* process is a process with no exchange of heat with the surroundings.
- An *isothermal* process is a process carried out at constant temperature.
- An *isobaric* process is a process carried out at constant pressure.
- An *isochoric* process is a process carried out at constant volume.
- An *exothermic* (or *exothermal*) process is a process that evolves heat. The opposite is an *endothermic* (or *endothermal*) process that consumes heat.

Heat and work

The most common way the system interacts with its surroundings is by exchange of *heat* and *work*. Heat and work are both energy in transition i.e., energy that is transferred.

If the system is heated with energy from the surroundings, we say that heat enters the system or that heat is transferred to the system. The temperature increases as the system accumulates more energy. We say that the system increases its thermal energy, i.e., energy in the form of molecular motion and vibrations. It does not *contain* the transferred heat; it contains thermal energy that was transferred as heat. Heat as well as thermal energy are disordered forms of energy as the energy does not act in a specific direction.

The other form of energy in transition is *work*. In contrast to heat, it is directional and ordered. If the system is compressed from the outside, work is exerted on it. This is called compression work and is often visualized by a piston moving in one direction. The definition of work is that it is *movement against an opposing force*. In the moving piston case, the opposing force is that of the pressure in front of the piston. After compression, the system has increased its energy because the molecules in the system have been accelerated by the piston. Gaseous molecules that constantly bombard the piston leave the piston surface with a slightly higher kinetic energy because they were pushed while bouncing

off. The system now contains the compression energy as an increase in thermal energy. It does *not* contain the work applied to it. Moreover, one cannot afterwards tell from the system if the increase of thermal energy was caused by a flow of heat or by work. Examples of other forms of work are stirring motion (locally ordered to begin with) and electrical energy in a wire (also directional to begin with).

The thermodynamic state functions

The fundamental thermodynamic energy functions are the *internal energy*, the *enthalpy*, the *Gibbs free energy* and the *Helmholtz free energy*. The last fundamental thermodynamic function is the *entropy*. This is not an energy function since it has another dimension (another unit). All the mentioned functions are *state functions*. Ther term state function means that it determined by the state of the system (volume, pressure, temperature and composition). If the state of the system is given, its thermodynamic functions are given too. In other words, two identical system also have identical thermodynamic functions.

In the following, the thermodynamic functions will be summarized. Heat and work are not state functions since they are not defined by the state of the system, only by the process in which they are transferred. One could say that they are immediately "forgotten" after being transferred.

Internal energy, U

The internal energy, *U*, is in principle all the energy the system contains, regardless the form. It includes thermal energy, chemical energy, any kinetic or potential energy and even nuclear energy. We cannot easily determine the total internal energy of a system, but that is not a problem either, because we are only interested in the changes, and changes can be accounted for with high accuracy. These changes are changes in thermal energy brought about by heat and work and in some cases by chemical reactions evolving or consuming heat[1]. For a system that is only changed by flows of heat or by work (and if no chemical reactions take place), we need not consider the chemical energy of the system, since it doesn't change. Likewise, nuclear energy is normally not considered either.

So, practically in thermodynamics, the internal energy to consider is the thermal energy of the system, which is contained by the molecules and atoms by their motion, vibrations and repulsive and attractive interaction (kinetic and potential energies). Again, this is called *heat* in colloquial language, but not in thermodynamics. In thermodynamics, thermal motion of the molecules in the system is indeed *thermal energy*.

Enthalpy, H

The enthalpy, *H*, is a slightly modified form of internal energy. By definition, it is the internal energy with addition of the product of the volume and the pressure of the system.

$$H \equiv U + pV \tag{1}$$

[1] The use of the term *heat* here is justified by the fact that heat from the chemical reaction is added to the thermal energy pool of the system (or consumed by the chemical reaction)

Enthalpy is more practical to use than internal energy in systems of constant pressure. When the internal energy changes, the system volume will change too to maintain constant pressure. This implies exchange of pressure-volume work with the surroundings and thus an accompanying additional change of internal energy, which complicates calculations. When using enthalpies, the exchange of work with the surroundings that the volume change implied is exactly compensated by the change in the *pV* term of the enthalpy. In other words, the *pV* term is a buffer always keeping track of the work balance. Like for internal energy, we are only interested in the changes, and true absolute values are irrelevant. The absolute values found in tables are based on a given definition of the starting point of the scale, just like for temperature, which can be quantified in centigrade and Kelvin with the only difference being the zero of the scales. A simplified, yet useful statement is that enthalpy is the practical internal energy at constant pressure. Like the internal energy, the enthalpy of a perfect gas is independent on pressure.[2]

Entropy

Entropy, *S*, is referred to as the degree of disorder in the system, but more correctly, it is a measure of the spreading of energy and matter. It is first of all a statistical phenomenon. The case where thermal energy is spread among all molecules present is more likely than a case where it is concentrated in one area or among a small number of molecules. Likewise, it is much more likely that molecules in a gas are spread over the entire volume available than that they are accumulated in certain locations. The total entropy thus has to components, the *thermal entropy* (Spreading of the energy among the molecules), and *spatial entropy* (spreading of the molecules in space).

Entropy is easily created in a process, but it cannot be destroyed. This is the key to the importance of entropy in relation to energy technologies. No single process can make the energy less disordered or less spread, only the opposite. A system can indeed change state to a less disordered one (i.e., with lower entropy), but only if the process of change is coupled to another process elsewhere leading to a gain of entropy of at least the same magnitude.

The *second law of thermodynamics* is also called the entropy law. It states that entropy cannot be destroyed. Instead, it always increases. The concept of entropy is explained in more detail the box below: "*Elaboration on entropy*"

It is a good rule of thumb that with a fixed amount of matter, the entropy of a gas phase is much higher than that of a condensed phase (a liquid or a solid) because the gas molecules spread over the entire volume available while condensed matter only occupies its own specific volume. A liquid has higher entropy that a crystalline solid since the liquid is less ordered with no fixed position in contrast to the crystalline solid. Entropy is ranked in the following order:

$$S_{gas} >>> S_{liquid} > S_{solid} \quad (2)$$

If a chemical process leads to a net increase of gaseous molecules, one can be very certain that the entropy change of the reaction is positive. Likewise, if there is a net reduction in the number of gaseous molecules, entropy diminishes. This is not allowed, but the process might still be possible if it generates

[2] The reason for this is that in a perfect gas the thermal energy is contained as molecular movement only. Since the molecules do not sense each other, the energy does not depend on how close they fly together. The pressure does not contain energy, but it is important if the gas shall perform work by expansion. Expansion is driven by pressure, but the energy is contained in the molecules.

heat. When calculating the change in thermodynamic functions in a reaction, constant temperature is assumed. This means that a heat evolving reaction (an exothermic reaction) supplies heat to the surroundings to maintain a constant temperature. This heat flow carries entropy and since the thermal energy of the flow is generated by the process, the entropy of it is generated too. This generated thermal entropy is often much higher than the lost spatial entropy (spreading in space) of the reaction and thus the second law is not violated. This balancing principle is the basis for concept of Gibbs free energy.

Elaboration on entropy

Entropy is defined statistically by Boltzmann's entropy formula

$$S = k_{\mathrm{B}} \ln W \tag{3}$$

Where *S* is the entropy of the system, k_{B} is Boltzmann's constant and *W* is the number of different ways the system can be made, each with different positions of the molecules or atoms and each with a different distribution of the energy among the molecule and atoms. It is obvious that the number, *W*, is always extremely large when we talk molecules and atoms in a system. We call all these ways to make the same system *microstates*. Each of the microstates compose the same *macrostate*, which is the state we can "see" in our macroscopic world, defined by composition, temperature and pressure. The theory then assumes that all microstates are equally possible and consequently, if two different macrostates are possible, then the one with the most microstates is most probable and will be the one that exists.

Boltzmann's formula tells us clearly that the macrostate with the highest number of microstates also has the highest entropy, and that it will be the one we see. The outcome of any process is that with the highest entropy simply for statistical reasons. However, the formula is not very practical for calculations unless in statistical mechanics. Calculations in thermodynamics are therefore made by the thermodynamic definition of entropy by *Clausius' inequality* (4):

$$\mathrm{d}S \geq \frac{\mathrm{d}Q}{T} \tag{4}$$

With Clausius' inequality we can determine the change of entropy, d*S*, by a flow of heat, d*Q*, into the system. *T* is the temperature of the system. The inequality says that when the system is heated so much, the entropy of it increases equivalently or more. No very exact, but the explanation follows.

The entropy of the system can change in two ways. It can change due to the entropy that the flow of heat carries (the spread of the energy in the heat flow). Then entropy is only transferred, and no new entropy is generated. The heat transfer is then *reversible* and the statement of (4) changes from "≥" to "=" (5).

$$\mathrm{d}S = \frac{\mathrm{d}Q_{\mathrm{rev}}}{T} \tag{5}$$

In general, a reversible process is a process that does not produce entropy. When it doesn't increase entropy, the reverse process does not either and therefore the process can proceed forward or

backward equally easily without violating the second law (saying that entropy cannot be destroyed). It is reversible. The reversible case is the case when heat is transferred without a temperature difference (an infinitively slow thermodynamic extreme, since the driving force for heat transfer is indeed a temperature difference). If heat is transferred across a temperature difference, as in the real world (e.g., lukewarm water is heat by a warm hotplate), the entropy increases additionally by generation of entropy as the concentrated energy of the high-temperature inflowing heat is dissipated among the molecules of the system which are originally at a lower temperature and thus a lower thermal energy level.

To illustrate this, we can imagine a high-temperature source of heat and a low temperature heat sink (receiving the heat). Heat leaves the hot source at T_h (thus reversibly) and the loss of entropy in the source is calculated by (5) (negative dQ gives negative dS). The same amount of heat enters the cold sink reversibly at T_c, and dS is likewise calculated by (5). Because $T_h > T_c$, the entropy gain in the cold sink is larger than the entropy loss in the hot source. Thus, entropy must have been generated on the way from the source to the sink.

As the change of entropy is positive, the process is possible (allowed by the second law). The opposite process, on the other hand would cause destruction of entropy, which is not allowed and will not happen. The latter is in perfect agreement with our experience, that heat only flows spontaneous from hot to cold and never the other way.

A good argument against this calculation could be to question our right to claim that heat leaves the hot source reversibly and enters the cold sin reversibly. The answer is that if the border of the hot system is just within the hot system, then the hot system cannot tell the temperature just outside the system and flow of heat would look exactly the same no matter the temperature outside. Outside, the temperature drops to the temperature of the cold sink over the distance between them, and this is where the entropy is generated. We may shorten that distance and bring source and sink infinitely close to each other without changing anything in the argumentation and then the limit is a direct contact.

In real systems, heat transport always requires a temperature difference, but it can be small if we have time and patience. Thermodynamics has that, and it is not a problem with infinitely slow processes.

In a real system with heat transfer between a hot source and a cold sink, local cooling will take place in the hot source near the exit. In principle a more sophisticated calculation could accommodate that, but the result in the end would be exactly the same as in our idealized case. The reason is that entropy is a state function. If we remove so much heat from the source the new state at a slightly lower temperature is given regardless of how we get to it. The same arguments count for the cold sink.

We can use (5) to calculate changes in entropy from heat transfer, but we can also use it for volume change. The trick is that since entropy is a state function, we can choose any path from the initial state to the final state. We can do the volume change by reversible compression or expansion processes and count any necessity heat flow in or out of the system. Reversible processes do not generate entropy, so the only entropy change (by transport) is via the reversible heat flow needed to reach the correct final state.

Entropy is not an energy function. It can be seen from (4) that the unit of entropy is J K^{-1} for absolute quantities or J K^{-1} mol^{-1} or J K^{-1} kg^{-1} for molar or mass-specific quantities.

Gibbs free energy

The ultimate indicator deciding whether a process will happen or not is the total (universal) change of entropy resulting from it (including all other coupled processes). A common coupled process is the transfer of heat to the surroundings from an exothermic (heat evolving) process, which could be the combustion of a fuel. If the fuel is hydrogen the reaction

$$H_2(g) + ½O_2(g) \longrightarrow H_2O(l) \quad (6)$$

leads to a net loss of 1.5 moles of gas molecules per mole of reaction. The spatial entropy is thus reduced. At the same time, heat is transferred to the surrounding since the process is highly exothermal (heat evolving). The thermodynamic driver of the process is the total entropy balance, not what is given by the change of entropy with all species at the same temperature.

The reaction heat is the enthalpy change of the reaction, ΔH_r. It is the total energy liberated by the reaction. If the fuel is combusted for the purpose of heating, ΔH_r is total heating energy. If the fuel is combusted for the purpose of generating work (mechanical or electrical work), we must consider the entropy balance. If the process (reaction (6) or any other combustion process) leads to a loss of entropy, then we cannot convert all the reaction enthalpy to work. Work carries not entropy, since it is ordered, and then overall, we destroy entropy, which is not possible.

The fix is to use some of the evolved energy to dissipate as heat in the surroundings and this way generate the missing entropy there. We compensate for the lost spatial entropy of the reaction by generating thermal entropy in the surroundings. The minimum heat that must be wasted and dissipated in the surroundings is $Q_{\min}$

$$Q_{\min} = T\Delta S_r \quad (7)$$

Where ΔS_r is the entropy change of the reaction at the temperature *T*. Justification follows in the box below. If the reaction enthalpy is used for heating, the generation of entropy is normally more than sufficient to "cover the loss".

Coming back to the generation of work, we must "waste" $T\Delta S_r$ as a part of the process. The remaining energy that can be used freely, e.g., for generating work, is the Gibbs free energy of the reaction, ΔG_r. Thus, it follows that

$$\Delta G_r = \Delta H_r - T\Delta S_r \quad (8)$$

The Gibbs free energy is the maximum work that can be generated form a chemical process. The same argumentation goes for any process, chemical or mechanical and the general term describing the maximum generation of work possible is *exergy*, but it is beyond the scope here to go into that.

Justification of: $Q_{\min} = T\Delta S_{\mathrm{r}}$

This justification builds on equation (5) in the box elaborating in entropy. It is repeated here

$$\mathrm{d}S = \frac{\mathrm{d}Q_{\mathrm{rev}}}{T} \tag{9}$$

If T is the temperature of the surroundings, then dS is the entropy generated if the heat dQ_{rev} is transferred reversibly and dissipated in the surrounding. Let dS equal the lost entropy of the reaction numerically. Then dQ_{rev} is the minimum energy that must be wasted to balance the lost entropy. Rearranging (9) gives

$$\mathrm{d}S_{\mathrm{r}} = \mathrm{d}S = \frac{\mathrm{d}Q_{\mathrm{rev}}}{T} \quad \text{or} \quad \mathrm{d}Q_{\mathrm{rev}} = T\mathrm{d}S_{\mathrm{r}} \tag{10}$$

With negative $\mathrm{d}S_{\mathrm{r}}$, $\mathrm{d}Q_{\mathrm{rev,min}}$ becomes negative too seen form the system. This makes sense since the heat leaves. Integration of (10)

$$\int_{0}^{\mathrm{d}Q_{\mathrm{rev,min}}} \mathrm{d}Q_{\mathrm{rev}} = \int_{S_1}^{S_2} T\mathrm{d}S_{\mathrm{r}} \tag{11}$$

Gives

$$Q_{\mathrm{rev,min}} = T\Delta S_{\mathrm{r}} \tag{12}$$

Note that energies leaving the system are signed negative. This means that work or heat delivered by the system are signed negative. An exothermic reaction has a negative value of ΔH_{r} and if it is capable of producing work, the value of ΔG_{r} is also negative. Consequently, a negative change of entropy of the reaction causes a less negative value of ΔG_{r} and less work can be delivered by the system.

If the process results in a net increase of entropy (at constant temperature) the capability to deliver work exceeds the enthalpy of reaction. This is possible, but to observe energy conservation, only if heat can be absorbed from elsewhere.

Helmholtz free energy is similar to Gibbs free energy but defined at constant volume instead of constant pressure. It relates to internal energy as Gibbs free energy relates to enthalpy. I most cases, only enthalpies and Gibbs free energies are considered.

Summary of the thermodynamic functions

The internal energy, U, is the true content of energy in the system.

The enthalpy, H, is the practical internal energy used in most calculations. ΔH_r is the total heat that can be extracted form a process. A negative value means the energy can be extracted from the system.

The entropy, S, is the degree of spreading or disorder in the system. The total ΔS_r of all coupled processes cannot be negative.

The Gibbs free energy ΔG_r of a process (or just the free energy) is the maximum work obtainable from the process. It must be negative if work is to be delivered to the outside.

Thermal engines

The Carnot limitation

Any heat engine, i.e., a heat driven machine that produces work from a flow of heat, is ultimately limited in conversion efficiency by Carnot's principle stating that the efficiency, η_{max}, of the conversion process depends on the temperature of the heat source, T_h, and the temperature at which unutilized heat is liberated from the system, T_c.

$$\eta_{\max} = \frac{T_{\mathrm{h}} - T_{\mathrm{c}}}{T_{\mathrm{h}}} \tag{13}$$

The Carnot limit is restricting any process in which heat is converted to work. Examples are internal combustion engines, thermal power plants, steam engines, jet engines and Stirling engines. Real engines have even lower efficiency, due to various imperfections (e.g., heat loss, friction and unideal combustion), and by reducing the imperfections, the Carnot limit can be approached but never met and certainly not passed. Fuel cells, which operate by a more direct energy conversion process without converting the chemical energy into heat first, are not limited by Carnot's principle, but still limited by the change in Gibbs free energy of the fuel consumption process. The working principle of a Carnot machine is shown for a piston system in Figure 2. The same principles apply to turbine in a power plant where hot vapour at high pressure pushes the turbine blades. The work transferred to the blades cools the vapour and colder or perhaps condensed vapour leaves the turbine in the other end. Again, a flow with two temperatures. In a combustion engine, heat and pressure are generated inside the piston by a combustion process, the piston is moved, and the combustion gas is cooled by the work transferred out. Finally, it leaves during the exhaust stroke at a lower temperature. Again, two temperatures.

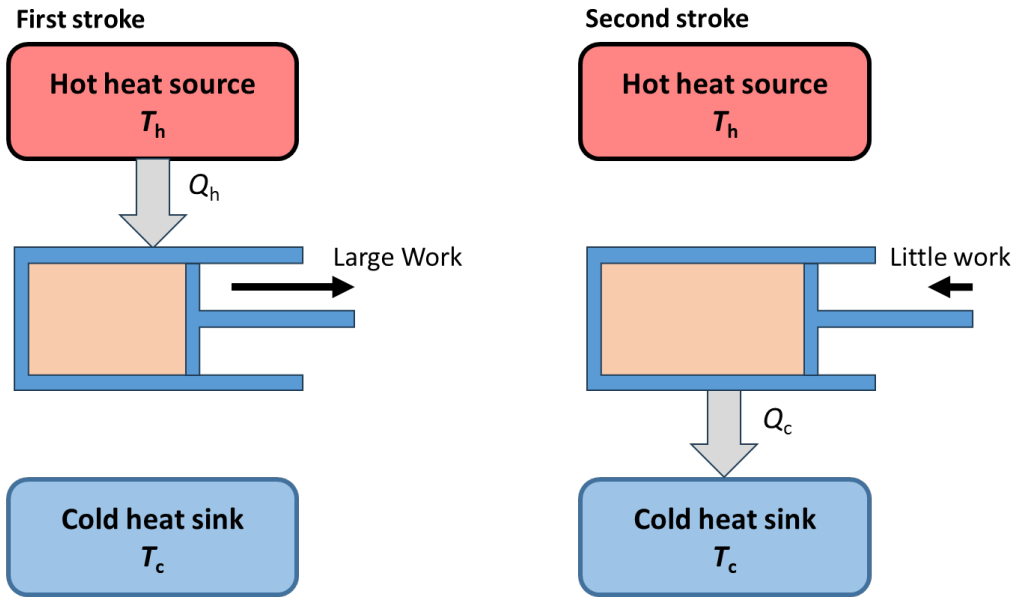


Figure 2. The principles of a thermal engine. In the first stroke, heat from a hot source is transferred to a medium confined in a piston system. This heats the medium and the pressure increases. At the same time the increased pressure moves the piston out, and work is generated (remember, work is movement against an opposing force. The opposing force is caused by the system utilizing the work (not shown)). The generation of work causes a flow of energy out of the system and this process cools the medium. Work is in this way generated from a flow of heat, but only until the piston reaches its extreme position. Before more work can be generated, the piston must reverse. This happens in the second stroke when the medium is cooled by the cold heat sink and contracts. The piston is pushed back by the pressure of the atmosphere, and a limited amount of work is given back to the system. The heating caused by that is counterbalanced by cooling by the cold sink. The system is now ready for another round. It is obvious that the system cannot work on the hot flow alone. The cold flow is just as important for closing the cycle.

Carnot's principle can be justified by considering the entropy balance of the conversion process. See Figure 3 and the justification box. A flow of heat from high temperature to low temperature without generation of work is spontaneous because the cold heat sink gains more entropy than the hot heat source loses. In both ends $\Delta S = Q/T$, in the ideal case with same heat flow, a low temperature gives the highest entropy. Thus overall, entropy increases. If part of the heat energy is converted into work by the machine (we don't care how), the flow of heat into the cold sink is less, but there can still be a net increase in entropy because heat enters at a lower temperature and each joule generates more entropy. The limiting case is when the entropy is just kept constant with balanced entropy flows out of the source and into the sink.

More intuitively, when work is generated from heat, it is done by thermal expansion of a medium and to close the cycle, the medium must be contracted again before it can be re-expanded. This contraction is driven by heat liberation at a lower temperature, T_{low}. See Figure 2 again.

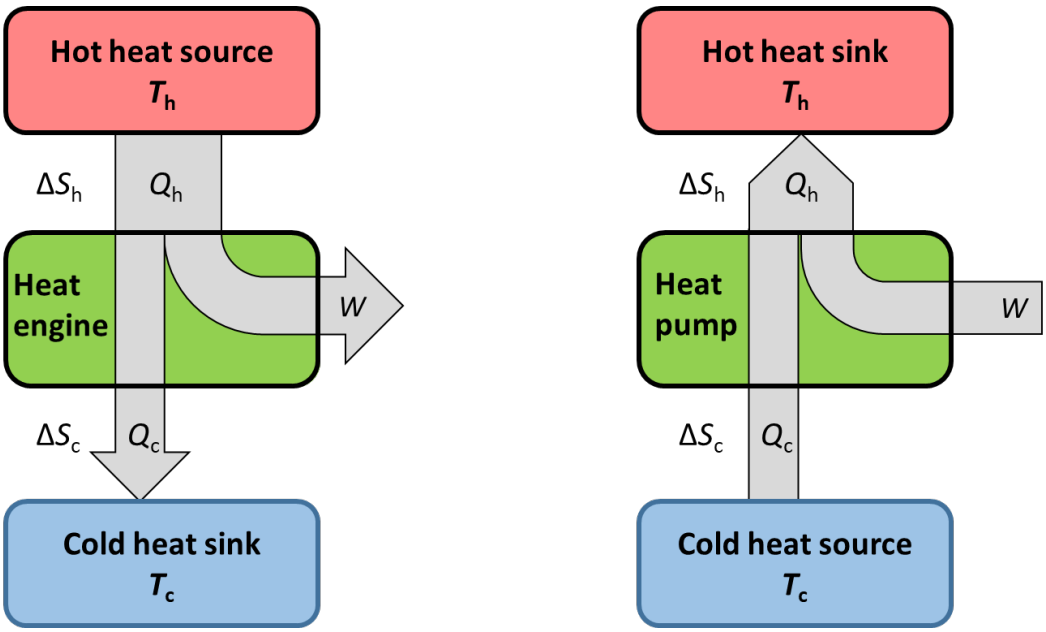


Figure 3. Thermodynamic principles of a heat engine (left) and a heat pump (right). The grey energy flows show that energy is conserved. In the heat engine the thermal energy is divided into work and remaining low grade thermal energy. In the heat pump the input of electrical energy (work) is topped up by low grade thermal energy. Judging from the width of the energy flows the COP[3] of the Heat pump is about 2.

Justification of Carnot's law (13), $\eta_{\max} = \frac{T_h - T_c}{T_h}$

Due to the conservation of energy, the work generated and delivered, *W*, equals the heat energy consumed

$$W = Q_h - Q_c \tag{14}$$

In the ideal case no heat is wasted for unnecessary entropy production, but the total entropy must be maintained in the system. The entropy balance with maximum work generation is (without signing)

$$\Delta S_h = \Delta S_c \quad \text{or} \quad \frac{Q_h}{T_h} = \frac{Q_c}{T_c} \tag{15}$$

In the second equation in (15), we assume the ideal case with reversible heat transfer out of the hot source and into the cold sink.
Rearranging (15) and insertion into (14) gives

[3] Explained below.

$$W_{\max} = \left(1 - \frac{T_c}{T_h}\right) Q_h \tag{16}$$

Which corresponds to (13). The value in the brackets is the Carnot factor or Carnot efficiency, $\eta_{\max}$.

$$W_{\max} = \eta_{\max} Q_h \quad \text{with} \quad \eta_{\max} = 1 - \frac{T_c}{T_h} = \frac{T_h - T_c}{T_h} \tag{17}$$

One may also look at it this way: if efficiency is "what you get" divided by "what you pay", the temperature difference, $T_h - T_c$, in the numerator of the last term of (17) is kind of a measure of the energy extracted from the flow (the work obtained, what you get) and T_h in the denominator is a similar measure of all the high quality energy supplied from the hot source (the energy spend, what you pay).

Heat pumps

A heat pump is a device that upgrades heat to heat at a higher temperature. It is mostly used for domestic heating but can just as well be used in industrial processes. The idea is to utilize a low-grade heat source for heating something already being at a higher temperature. This process is not spontaneous but can be forced by work.

In a thermodynamic sense, a heat pump is the reverse of a heat engine. See Figure 3. Even in the ideal case described in the previous part, the efficiency of a heat engine is always below one, so we can expect that the efficiency defined similarly for the reverse process is then above one. The missing entropy in the flow into the *hot* sink is generated from the work applied to the device. Again, in the ideal case the entropies of the two heat flows are just balancing. The work contains no entropy but generates entropy in the device. As we often see in thermodynamics, it is not important how such a heat pump is constructed, we only set the limits that thermodynamics dictates.

A real physical heat pump can be constructed in different ways, but the usual way is shown in Figure 4. A liquid with a suitable boiling point is evaporated by low grade heat from the low temperature heat source. It is then compressed (this is where work is applied) and the compression work results in a heating of the vapour (remember, the energy added as work is contained as thermal energy. It is not contained in the pressure). Due to the increased pressure, the vapour condenses in the condenser, and the latent heat (condensation heat) heats up the fluid further and the heat, or part of it, is released at the high temperature. After releasing heat, the fluid, now in liquid form, passes a nozzle after which the pressure is much lower, and it evaporates again. This time the latent heat leads to a cooling of the fluid it meets the low temperature heat source and the circle is completed.

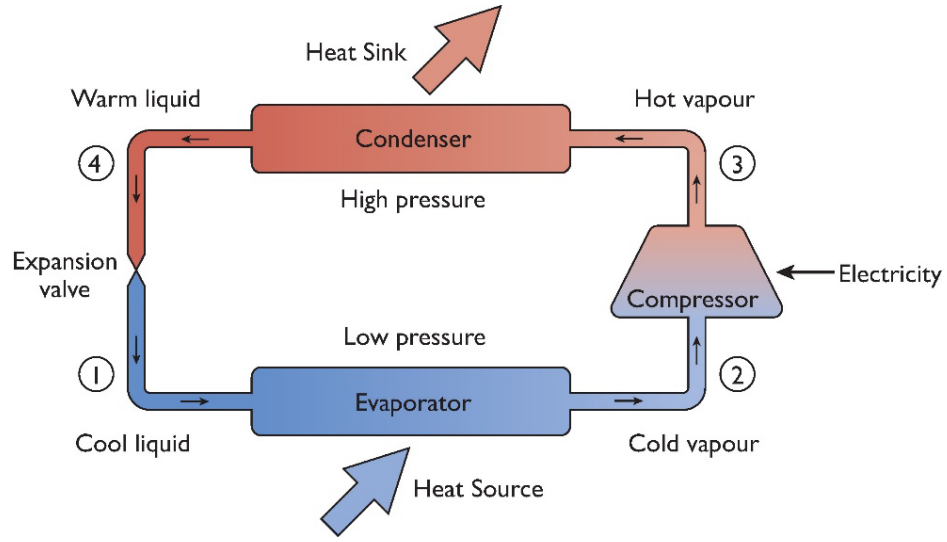


Figure 4. The usual working principle of a heat pump. 1, the expanded fluid evaporates and is cooled by the latent heat. It becomes cold enough to pick up heat from the low-temperature, low-grade, heat source. 2, the vapour is compressed and heats up. In the condenser it liberates high temperature (high grade) heat to the hot heat sink. During that process it cools gain and condenses liberating the latent heat. 4, finally, the cycle is closed when the high-pressure liquid passes through the expansion valve again.

When used for domestic heating, the low-grade heat source can be the air outside or underground pipes with a circulating heat transfer liquid. With efficiencies above one, heat pumps are normally not described by the term efficiency but by the term *Coefficient of Performance* (COP).

$$\mathrm{COP} = \frac{\dot{Q}_\mathrm{h}}{\dot{W}_\mathrm{comp}} = \frac{P_\mathrm{h}}{P_\mathrm{el}} \tag{18}$$

Where $\dot{Q}_\mathrm{h}$ is the heating heat flow, $\dot{W}_\mathrm{comp}$ the compressor workflow, P_h the heating power and P_el the electrical power used for the compressor. COP depends on the heat pump, the mode of operation and on the temperatures of the heat source and sink. Therefore, a heat pump cannot be labelled a specific COP but is characterized by charts with different conditions. However, common COP values are 2-4.

The ultimate idealized COP is

$$\mathrm{COP}_\mathrm{max} = \frac{Q_\mathrm{h}}{W_\mathrm{min}} \tag{19}$$

and can be found from Carnot's formula (13) slightly rearranged now with W_min instead of W_max. This is the same value for work, the limiting case.

$$W_\mathrm{min} = \frac{T_\mathrm{h} - T_\mathrm{c}}{T_\mathrm{h}} Q_\mathrm{h} \tag{20}$$

Inserting (20) into (19) gives

$$\mathrm{COP}_{\max} = \frac{T_\mathrm{h}}{T_\mathrm{h} - T_\mathrm{c}} \tag{21}$$

Intuitively, COP states how many times the desired hot heat flow is larger than the electrical energy causing it.

Thermodynamic functions of chemical reactions

When a chemical reaction takes place, thermal energy is either liberated or consumed as a result of the energy balance between chemical bonds formed and chemical bonds broken. The energy of a chemical bond is negative in the sense that when two atoms form a bond, they commonly attain a lower energy level than before (otherwise they would not bond). As energy cannot vanish it is liberated as thermal energy. In a total chemical reaction, some bonds are broken, and others are formed. If the chemical energy is lowered by the process, the liberated energy is concentred among the product molecules as thermal energy. If the energy would remain with the product molecules as kinetic energy in velocity and vibrations, the process could easily reverse regenerating the reactant molecules, because the necessary energy would still be close by and ready to be consumed again. However, the highly concentrated energy will shortly after be dissipated among other molecules and thus be less available to a reverse reaction. This spreading of the reaction energy increases the entropy of the system, and the likelihood of the reverse reaction is small.

It is very useful to determine the change in the thermodynamic functions as consequences of chemical reactions. In tables it is always deemed that the final temperature is the same as the initial temperature, very often 298.15 K, and that the pressure of all reactants and all products is the standard pressure, 1 bar. Any thermal energy generated is thus assumed transferred out of the system. This is the heat of reaction. If thermal energy is consumed by the process, the equivalent amount is assumed transferred to the system.

The change in any of the thermodynamic functions is then the sum of the respective values of the products (what is gained) minus that of the reactants (what is consumed or lost). In (23), *x* is the molar thermodynamic function of interest. Δx_r is the molar change of thermodynamic function of the reaction and x_i and x_j are the thermodynamic functions of the reacting and formed species. ν_i and ν_j are the stoichiometric coefficients, i.e., the numbers in the chemical equation stating the number of moles of each molecule (22). (*x* should here not be confused with molar fractions)

$$\nu_A\mathrm{A} + \nu_B\mathrm{B} + \cdots \rightarrow \nu_C\mathrm{C} + \nu_D\mathrm{D} + \cdots \tag{22}$$

$$\Delta x_r = \sum_{prod.} \nu_i\, x_i - \sum_{reac.} \nu_j\, x_j \tag{23}$$

This is the general principle used for both enthalpy, entropy and free energy of reaction and now we only need decide which molar values can be assigned to the involved species.

As stated above, we cannot easily (practically) determine absolute values of the enthalpy of a system or a substance, just as we cannot for the internal energy, but suitable definitions have been made. One valid way is just to decide on one arbitrary state as basis for the zero point of the scale for enthalpy, entropy and Gibbs free energy. This is an approach similar to the Celsius scale starting at the melting point of ice. It works since we are only interested in the changes, but a smarter and well adapted way is to use *enthalpies of formation* and *free energies of formation*. For the entropy scale, so called third law entropies are used. This is all explained in the following paragraphs.

Standard enthalpy of formation

Molar standard enthalpy of formation, $\Delta h_{\mathrm{f}}^{0}$, is defined as the enthalpy of formation of the compound by chemical reactions from the constituting elements in their stable states, i.e., at 1 bar and typically at 298.15 K. The "^{0}" indicates that standard conditions apply. Taking water as an example, the stable state at these conditions is liquid, not gas or solid. This is trivial, but many solids can exist in different structures and then the one with the lowest energy is considered the stable form. The enthalpy of formation is then the total enthalpy change of the reaction forming liquid water at 298.15 K from hydrogen gas and oxygen gas each at 1 bar and 298.15 K. The next question is of course how to determine the $\Delta h_{\mathrm{f}}^{0}$ of oxygen and hydrogen, but that is simple because forming hydrogen from elements equals forming hydrogen from hydrogen, involves no change. By this definition, the standard enthalpy of formation thus becomes zero for all elements in their stable state at 1 bar and 298.15 K, and then the problem of defining absolute enthalpies is surpassed. Any compound can be assigned a standard enthalpy of formation, and the enthalpy change of any chemical reaction can subsequently be calculated as the difference between the formed and the consumed species. Since the enthalpy of the elements as well as the final compound are state functions, it doesn't matter through which chemical routes formation for the final compound happens.

For the general reaction (eq. (22)), the molar standard enthalpy change of reaction, $\Delta h_{\mathrm{r}}^{0}$, is

$$\Delta h_{\mathrm{r}}^{0} = \nu_{\mathrm{C}}\Delta h_{\mathrm{f,C}}^{0} + \nu_{\mathrm{D}}\Delta h_{\mathrm{f,D}}^{0} + \cdots - \nu_{\mathrm{A}}\Delta h_{\mathrm{f,A}}^{0} - \nu_{\mathrm{B}}\Delta h_{\mathrm{f,B}}^{0} + \cdots \tag{24}$$

and in the general case

$$\Delta h_{\mathrm{r}}^{0} = \sum_{\mathrm{prod.}} \nu_{\mathrm{i}}\, \Delta h_{\mathrm{f,i}}^{0} - \sum_{\mathrm{reac.}} \nu_{\mathrm{j}}\, \Delta h_{\mathrm{f,j}}^{0} \tag{25}$$

Again, the standard enthalpy of formation of any pure element is zero, because formation of an element from itself involves no change at all. Standard enthalpies of formation can be found in tables.

Standard free energy of formation

Molar standard Gibbs free energy change of reaction, $\Delta g_{\mathrm{r}}^{0}$, is defined the same way as enthalpy of formation, just with free energy values. It follows that calculation of molar standard Gibbs free energy of reaction is done by the general formula

$$\Delta g_{\mathrm{r}}^{0} = \sum_{\mathrm{prod.}} \nu_{\mathrm{i}}\, \Delta g_{\mathrm{f,i}}^{0} - \sum_{\mathrm{react.}} \nu_{\mathrm{j}}\, \Delta g_{\mathrm{f,j}}^{0} \tag{26}$$

Like the standard enthalpy of formation, the standard Gibbs free energy of formation of any pure element is zero. Standard Gibbs free energies of formation can be found in tables.

Standard third law entropies

In contrast to the functions, internal energy and enthalpy, a point of zero *has* been defined for entropy. It is done by the third law of thermodynamics stating that the entropy of any perfect crystal at the absolute zero of the temperature scale (0 K) has zero entropy. From that, the entropy content can be calculated at any other temperature and form by rewriting (4) a little and integrating it from T = 0 K to the temperature of interest. If the pressure is maintained as the standard pressure, 1 bar, the value is the standard entropy and again, tabulated values are typically given for T = 298.15 K and the expression becomes

$$s^0 = \int_{T=0\ \mathrm{K}}^{T=298.15\mathrm{K},} \frac{c_\mathrm{p}(T)}{T}\,\mathrm{d}T \tag{27}$$

$c_\mathrm{p}(T)$ is the constant pressure heat capacity, which is a function of temperature. Entropy changes from phase changes (melting, boiling and structural changes) on the way must be included by adding the entropy from the latent heat via (4). The calculation requires knowledge of how the heat capacity changes with temperature, and it is not a common activity to perform. Instead, already tabulated values can be used.

The molar standard entropy change of reaction is then

$$\Delta s_\mathrm{r}^0 = \sum_{\text{prod.}} \nu_\mathrm{i}\,\Delta s_\mathrm{i}^0 - \sum_{\text{react.}} \nu_\mathrm{j}\,\Delta s_\mathrm{j}^0 \tag{28}$$

Enthalpy of combustion

The main descriptor of a fuel is its *heating value* or *calorific value*, which is the amount of heat that is liberated on combustion with oxygen. Fuels as we know them are all made from hydrogen, carbon, oxygen and nitrogen. In the combustion process, hydrogen and oxygen atoms are converted to water, carbon atoms to CO_2 and nitrogen atoms into dinitrogen molecules, N_2. The latter is the ideal case and depending on the combustion process, various nitrogen oxides (NOX) may be formed to some extent.

Gasoline is a complicated mixture of hydrocarbons, but is often illustrated by octane, C_8H_{18}, for simplicity. The combustion process is as follows

$$C_8H_{18} + 12.5O_2 \rightarrow 8CO_2 + 9H_2O \tag{29}$$

The standard enthalpy of combustion, Δh_c^0, is the standard enthalpy of the combustion reaction calculated by equation (25) as:

$$\Delta h^0_{c,C8H18} = \Delta h^0_{\mathrm{r}} = 8\Delta h^0_{\mathrm{f,CO2}} + 9\Delta h^0_{\mathrm{f,H2O(l)}} - \Delta h^0_{\mathrm{f,C8H18}} - 12.5\Delta h^0_{\mathrm{f,O2}} \quad (30)$$

With the table values $\Delta h^0_{\mathrm{f,CO2}} = -393.51\ \mathrm{kJ\ mol^{-1}}, \Delta h^0_{\mathrm{f,H2O(l)}} = -285.83\ \mathrm{kJ\ mol^{-1}}, \Delta h^0_{\mathrm{f,C8H18}} = -249.9\ \mathrm{kJ\ mol^{-1}}, \Delta h^0_{\mathrm{f,O2}} = 0.0\ \mathrm{kJ\ mol^{-1}}$, the standard enthalpy of combustion, Δh^0_{c}, becomes -5479.65 kJ mol^{-1}.

As stated above, the term "standard" states that all species are at standard pressure, i.e., 1 bar, and in their stable states at the given temperature, which is 298.15 K according to normal practice. This means that water is liquid ($H_2O(l)$). Δh^0_{c} is negative since heat is liberated from the system. Had it been positive, the substance of our attention would not be a fuel since it is a key property of a fuel that it liberates energy on combustion.

Higher and lower heating value

The *higher heating value* (HHV) of the fuel is defined as $-\Delta h^0_{\mathrm{c}}$ (the enthalpy of combustion signed positive). As Δh^0_{c} is defined based on the product water being liquid, this is also the case for the higher heating value of a fuel. An equivalent *lower heating value* (LHV) is defined the same way, only with water in the vapour (gas) state ($H_2O(g)$) having a slightly lower enthalpy of formation. The difference is equal to the enthalpy change between liquid and vapour, the enthalpy of vaporization. It follows that the highest amount of energy one can extract form combusting a fuel is given by the HHV. This is also the most fundamental quantity in thermodynamics. However, LHV is commonly used with fuels since the product water is often vented at vapour. Other names for HHV and LHV are *gross calorific value* (GCV) and *net calorific value* (NCV), respectively.

Maximum heat and work of a fuel

The maximum heat obtainable from a fuel equals $-\Delta h^0_{\mathrm{c}}$, numerically its higher heating value. The maximum work obtainable from the same fuel is given by the equivalent free energy of combustion with opposite signing, $-\Delta g^0_{\mathrm{c}}$ (again, in thermodynamics the signing is defined from the system's point of view, but heat or work obtainable is from the spectator's point of view, i.e., from the surroundings, thus with the opposite signing). The Gibbs free energy of reaction (29) is

$$\Delta g^0_{c,C8H18} = \Delta g^0_{\mathrm{r}} = 8\Delta g^0_{\mathrm{f,CO2}} + 9\Delta g^0_{\mathrm{f,H2O(l)}} - \Delta g^0_{\mathrm{f,C8H18}} - 12.5\Delta g^0_{\mathrm{f,O2}} \quad (31)$$

And the maximum work obtainable from of reaction (29) is $-\Delta g^0_{c,C8H18}$.